

1.6 Vector Inner Product



Core skills you will master in this section:

- ▶ Computing the inner product: multiply corresponding components and sum the products; the result is a scalar.
- ▶ Properties of the inner product: master basic algebraic properties such as commutativity, distributivity, and compatibility with scalar multiplication.
- ▶ Geometric meaning: the inner product equals the product of the two vectors' lengths and the cosine of the angle between them.
- ▶ Determining orthogonality: use a zero inner product to determine whether two vectors are orthogonal.
- ▶ Angle and similarity: the smaller the angle, the more similar the vectors' directions.

The vector inner product, also called the scalar product or dot product, is a core concept in linear algebra with wide applications in geometry, physics, and computer science. This section helps you understand the inner product from two perspectives: algebraic and geometric. From the algebraic perspective, the inner product is the sum of the products of two vectors' corresponding components; it satisfies commutativity, distributivity, and compatibility with scalar multiplication, and it can be used to compute a vector's length. From the geometric perspective, the inner product relates the two vectors' lengths to the cosine of the angle between them, which in turn lets us judge directional relationships between vectors, such as orthogonality. Later in this chapter we will explore applications of the vector inner product such as orthogonal projection.

Algebraic Perspective

Given column vectors of the same dimension $\mathbf{a}$ and $\mathbf{b}$

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$\mathbf{a}$ and $\mathbf{b}$ have an inner product equal to the sum of the products of corresponding components — a scalar — namely

$$\mathbf{a} \cdot \mathbf{b} = \langle \mathbf{a}, \mathbf{b} \rangle = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n = \sum_{i=1}^n a_i b_i$$

Here, Σ is the summation symbol; the $i = 1$ below Σ means the summation starts at index $i = 1$ and continues up to $i = n$.

$a_i b_i$ denote the two vectors' i -th components, multiplied together.

$\mathbf{a}$ and $\mathbf{b}$'s inner product can also be written as $\langle \mathbf{a}, \mathbf{b} \rangle$; this book generally uses a solid dot $\cdot$ as the operator for the vector inner product. Note that the $\cdot$ dot cannot be omitted.

For example, the scalar product (inner product, dot product) of the following two 3-dimensional vectors is

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = 1 \times 4 + 2 \times 5 + 3 \times 6 = 4 + 10 + 18 = 32$$

Code 1 shows how to compute the vector inner product with NumPy below, and verify the result by hand. The code is simple — annotate it yourself as you study it.

Code 1. Vector Inner Product |  LA_01_05_01.ipynb

```
## 初始化
import numpy as np

## 定义两个向量
a_vec = np.array([1, 2, 3])
b_vec = np.array([4, 5, 6])

## 计算内积
a_dot_b = np.dot(a_vec, b_vec)

## 验算
a_vec * b_vec
sum(a_vec * b_vec)
```

Common Properties of the Vector Inner Product

The vector inner product satisfies the following commutative, distributive, and scalar-multiplication compatibility properties:

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \mathbf{b} \cdot \mathbf{a} \\ \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c} \\ k(\mathbf{a} \cdot \mathbf{b}) &= (k\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (k\mathbf{b}) \end{aligned}$$

Here, the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ all have the same dimension; k is a scalar.

The inner product of vector $\mathbf{a}$ with itself is $\mathbf{a}$ squared — the sum of the squares of all its components:

$$\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 = a_1^2 + a_2^2 + \cdots + a_n^2$$

Clearly, $\mathbf{a} \cdot \mathbf{a} \geq 0$; if $\mathbf{a} \cdot \mathbf{a} = 0$, if and only if $\mathbf{a} = \mathbf{0}$.

If one of the vectors in the inner product is the all-ones vector, for example

$$\mathbf{a} \cdot \mathbf{1} = a_1 + a_2 + \cdots + a_n$$

the result of the equation above is equivalent to summing all the components of $\mathbf{a}$.

Geometric Perspective

The definition of the vector inner product looks especially simple, yet it holds rich geometric meaning.

From a geometric perspective, the inner product of vectors $\mathbf{a}$ and $\mathbf{b}$ equals the lengths of $\mathbf{a}$ and $\mathbf{b}$ multiplied together, times the cosine of the angle between them, namely

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

As shown in Figure 1 (a), $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$ are the vectors' lengths (magnitude, norm, L^2 norm, Euclidean norm); θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

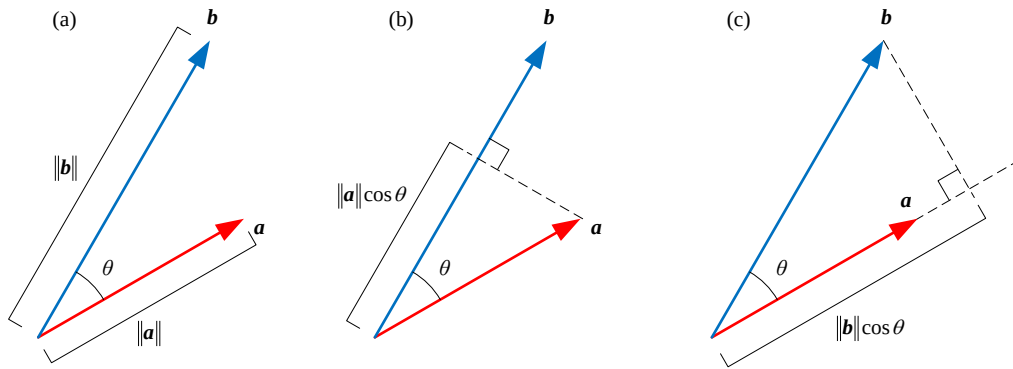


Figure 1. Geometric View of the Vector Inner Product

As shown in Figure 1 (b), $\|\mathbf{a}\| \cos \theta$ is equivalent to the “shadow” of vector $\mathbf{a}$ onto the direction of vector $\mathbf{b}$, we call this projection length the scalar projection. Later in this chapter we will introduce the scalar projection in detail.

Similarly, as shown in Figure 1 (c), $\|\mathbf{b}\| \cos \theta$ is equivalent to the scalar projection of vector $\mathbf{b}$ onto the direction of vector $\mathbf{a}$.

Angle

The equation above shows that a vector inner product depends not only on the vectors' lengths (norm, L^2 norm) but also on the cosine of the angle between them. As shown in Figure 2, given $\|\mathbf{a}\|$ and $\|\mathbf{b}\|$, $\|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$ varies with θ . Note that $\mathbf{a}$ and $\mathbf{b}$ are both nonzero vectors.

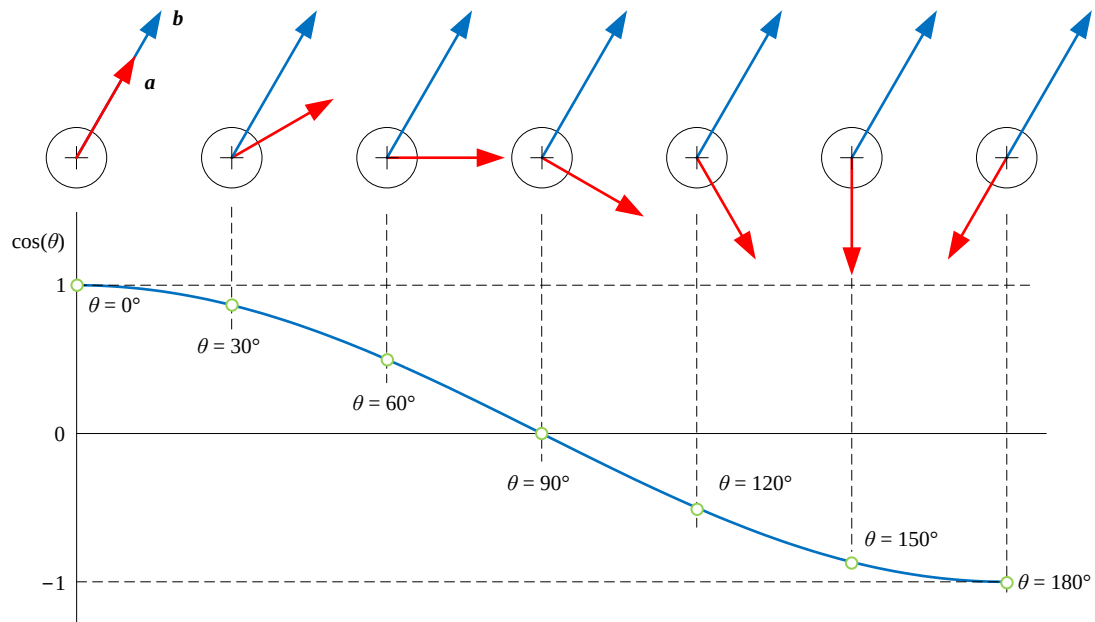


Figure 2. Vector Inner Product Changes with θ , While the Lengths of $\mathbf{a}$ and $\mathbf{b}$ (Norm, L^2 Norm) Stay Fixed

When $\theta = 0^\circ$, $\mathbf{a}$ and $\mathbf{b}$ point in the same direction, and the inner product is at its maximum ($\cos(0^\circ) = 1$).

When $0^\circ < \theta < 90^\circ$, $\mathbf{a}$ and $\mathbf{b}$ form an acute angle, and $\mathbf{a} \cdot \mathbf{b} > 0$.

When $\theta = 90^\circ$, $\mathbf{a}$ and $\mathbf{b}$ form a right angle, also called $\mathbf{a}$ and $\mathbf{b}$ being orthogonal, and $\mathbf{a} \cdot \mathbf{b} = 0$ ($\cos(90^\circ) = 0$); we call this type of vector set orthogonal vectors, meaning they are mutually orthogonal (though they need not be unit vectors).

When $90^\circ < \theta < 180^\circ$, $\mathbf{a}$ and $\mathbf{b}$ form an obtuse angle, and $\mathbf{a} \cdot \mathbf{b} < 0$.

When $\theta = 180^\circ$, $\mathbf{a}$ and $\mathbf{b}$ point in opposite directions, and the inner product is at its minimum ($\cos(180^\circ) = -1$).

If $\mathbf{a}$ and $\mathbf{b}$ are both nonzero vectors, rearranging gives

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

This can also be written as

$$\cos \theta = \frac{\mathbf{a}}{\|\mathbf{a}\|} \cdot \frac{\mathbf{b}}{\|\mathbf{b}\|}$$

Here, $\frac{\mathbf{a}}{\|\mathbf{a}\|}$ and $\frac{\mathbf{b}}{\|\mathbf{b}\|}$ are, respectively, the direction vectors (unit vectors) of $\mathbf{a}$ and $\mathbf{b}$. Since $\cos \theta$ always lies in $[-1, 1]$, this also gives the Cauchy–Schwarz inequality, $|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$, with equality exactly when the two vectors are parallel.

Code 2 computes the angle between two vectors. Let's walk through the key lines below.

a `numpy.array()` is used to define two vectors, `vector1` and `vector2`.

b `numpy.dot(vector1, vector2)` computes the inner product of the two vectors. The `np.dot(a, b)` function computes the dot product of two vectors – that is, it multiplies their corresponding elements and sums the results.

c `numpy.linalg.norm()` computes the lengths (norm, L2 norm, Euclidean norm) of `vector1` and `vector2`.

d Dividing the inner product by the product of the two vectors' lengths gives the cosine of the angle between them.

e `numpy.arccos()` computes the angle in radians.

f `numpy.degrees()` converts the angle from radians to degrees.

Code 2. Computing the Angle Between Two Vectors |  LA_01_05_02.ipynb

```
## 初始化
import numpy as np

## 定义两个向量
a vector1 = np.array([1, 1, 0])
    vector2 = np.array([0, 1, 1])

## 计算内积
b dot_product = np.dot(vector1, vector2)

## 计算向量长度
c norm1 = np.linalg.norm(vector1)
    norm2 = np.linalg.norm(vector2)

## 计算夹角的余弦值
d cos_theta = dot_product / (norm1 * norm2)

## 计算夹角（弧度）
e theta_rad = np.arccos(cos_theta)

## 计算夹角（角度）
f theta_deg = np.degrees(theta_rad)
```

Relationships Between Color Vectors in RGB

Next, let's use the vector inner product to analyze the relationships between color vectors in RGB space.

In RGB space, the red vector $\mathbf{e}_1$, green vector $\mathbf{e}_2$, and blue vector $\mathbf{e}_3$ have pairwise inner products of 0:

$$\mathbf{e}_1 \cdot \mathbf{e}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = 0, \quad \mathbf{e}_1 \cdot \mathbf{e}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0, \quad \mathbf{e}_2 \cdot \mathbf{e}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = 0$$

This shows that the red vector $\mathbf{e}_1$, green vector $\mathbf{e}_2$, and blue vector $\mathbf{e}_3$ are pairwise orthogonal – that is, the angle between any two of them is 90° . In other words, $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are orthogonal vectors.

What's more, $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ are all unit vectors; we call such a set of vectors an orthonormal set, meaning a group of vectors that are mutually orthogonal and each has unit length.

In other words, given an orthonormal set of vectors, such as $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$, every vector in the set is orthogonal to the others, and every vector has length 1.

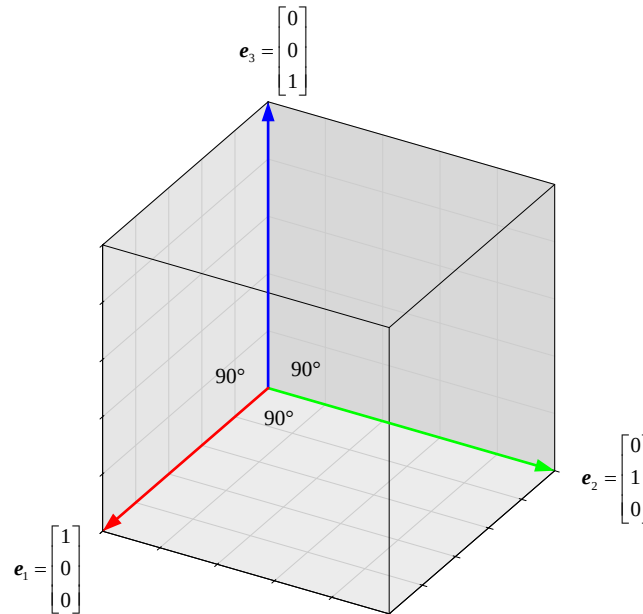


Figure 3. Orthonormal Vectors in RGB Space

When two color vectors have an inner product of zero, it means the two colors' components are completely independent, with no shared color component.

For example, in Figure 4, the color vectors orthogonal to the red vector include green, blue, and also cyan.

Cyan is formed by a linear combination of blue and green; put another way, any color vector formed by a linear combination of blue and green is orthogonal to red. In other words, the red vector is perpendicular to the entire "blue-green" plane.

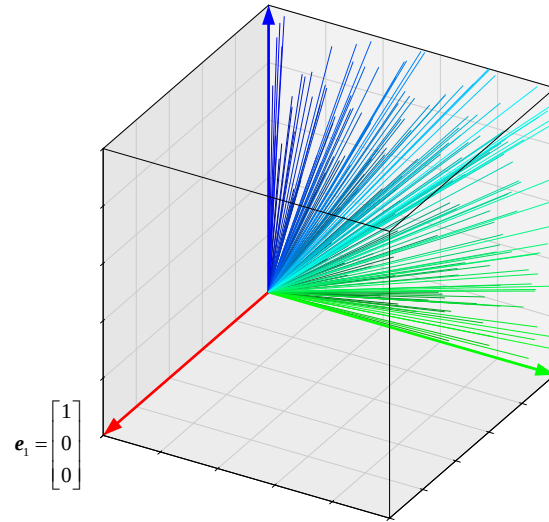


Figure 4. In RGB Space, the Red Vector Is Orthogonal to the Entire “Blue-Green” Plane

Similarly, as shown in Figure 5 (a), the blue vector is perpendicular to the entire “red-green” plane; as shown in Figure 5 (b), the green vector is perpendicular to the entire “red-blue” plane.

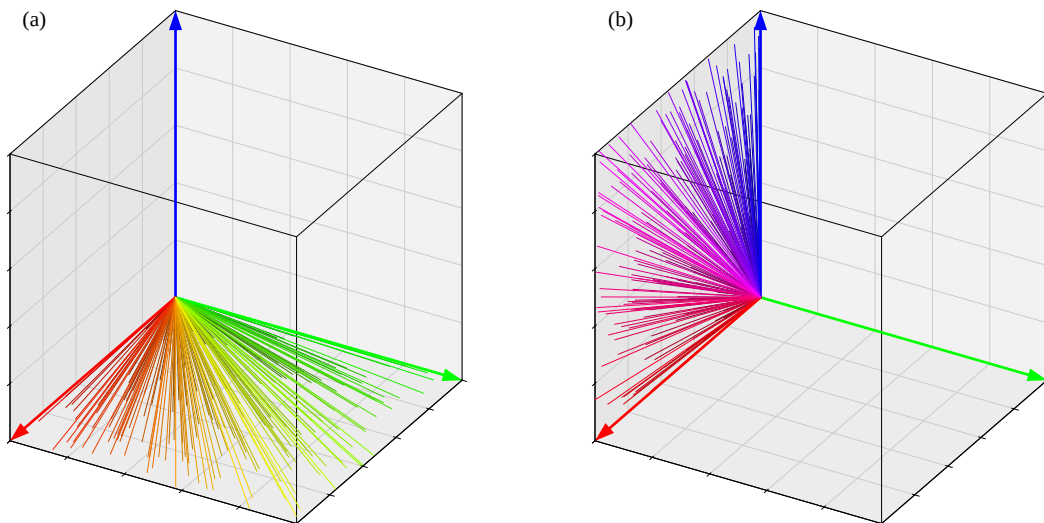


Figure 5. In RGB Space, the Blue Vector Is Perpendicular to the Entire “Red-Green” Plane, and the Green Vector Is Perpendicular to the Entire “Red-Blue” Plane

The yellow vector is formed by adding the red and green vectors together; clearly, the red and yellow vectors are no longer orthogonal.

The inner product of the red vector e_1 and the yellow vector $e_1 + e_2$ is

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 1$$

The red vector has length 1; the yellow vector has length $\sqrt{2}$. The cosine of the angle between the red and yellow vectors is

$$\frac{1}{\sqrt{1^2 + 0^2 + 0^2} \times \sqrt{1^2 + 1^2 + 0^2}} = \frac{1}{1 \times \sqrt{2}} = \frac{\sqrt{2}}{2}$$

which lets us compute the angle between the red and yellow vectors as

$$\arccos\left(\frac{\sqrt{2}}{2}\right) = 45^\circ$$

This shows that although the two colors are different, yellow contains a certain amount of red, so the two colors overlap.

As an exercise, compute the angle between the green and yellow vectors.

Also, think about which other pairs of vectors in Figure 6 have a 45° angle between them.

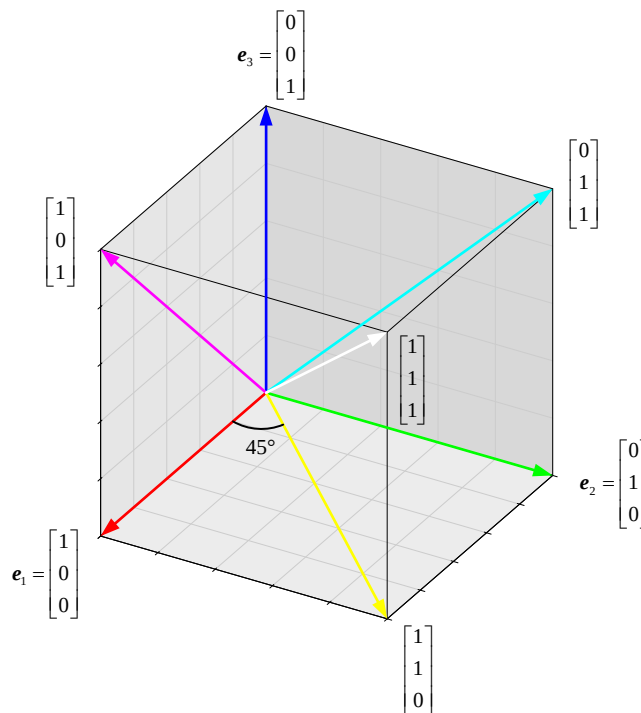


Figure 6. A 45° Angle Between Color Vectors in RGB Space

As shown in Figure 7, in RGB space, yellow, magenta, and cyan form a regular tetrahedron together with white. Each face of the tetrahedron is an equilateral triangle.

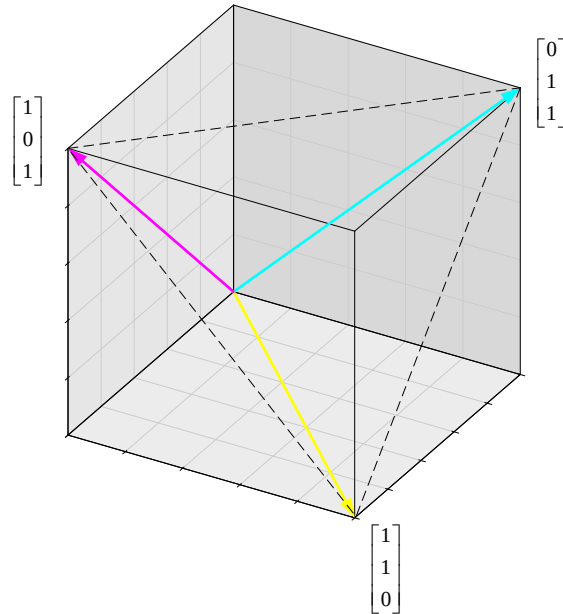


Figure 7. In RGB Space, White, Yellow, Magenta, and Cyan Form a Regular Tetrahedron

Now let's compute the inner product of the yellow vector $\mathbf{e}_1 + \mathbf{e}_2$ and the cyan vector $\mathbf{e}_2 + \mathbf{e}_3$:

$$\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 1$$

The cosine of the angle between the yellow and cyan vectors is

$$\frac{1}{\sqrt{1^2 + 1^2 + 0^2} \times \sqrt{0^2 + 1^2 + 1^2}} = \frac{1}{2}$$

The angle between the yellow and cyan vectors is

$$\arccos\left(\frac{1}{2}\right) = 60^\circ$$

Think about which other pairs of vectors in Figure 6 have a 60° angle between them.

Now let's compute the inner product of the white vector $\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$ and the red vector $\mathbf{e}_1$:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = 1$$

The cosine of the angle between the white and red vectors is

$$\frac{1}{\sqrt{1^2 + 1^2 + 1^2} \times \sqrt{1^2 + 0^2 + 0^2}} = \frac{\sqrt{3}}{3}$$

The angle between the white and red vectors is

$$\arccos\left(\frac{\sqrt{3}}{3}\right) \approx 54.8^\circ$$

As an exercise, compute the angle between the white and green vectors on your own.

Now let's compute the inner product of the white vector $\mathbf{e}_1 + \mathbf{e}_2 + \mathbf{e}_3$ and the yellow vector $\mathbf{e}_1 + \mathbf{e}_2$:

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 2$$

The cosine of the angle between the white and yellow vectors is

$$\frac{2}{\sqrt{1^2+1^2+1^2} \times \sqrt{1^2+1^2+0^2}} = \frac{\sqrt{6}}{3}$$

The angle between the white and yellow vectors is

$$\arccos\left(\frac{\sqrt{6}}{3}\right) \approx 35.3^\circ$$

As an exercise, compute the angle between the white and cyan vectors on your own.

When the angle between two color vectors is greater than 0° and less than 90° , it means the two colors share overlapping components — “each contains a bit of the other.” If the angle is 90° , the two color vectors are orthogonal, meaning there is no overlapping component at all.

As shown in Figure 8, the closer the angle is to 0° , the more similar the two color vectors' directions are, meaning the colors are closer together with a higher degree of overlap; the closer the angle is to 90° , the greater the difference in the two color vectors' directions, meaning the colors are more distinguishable with a lower degree of overlap.

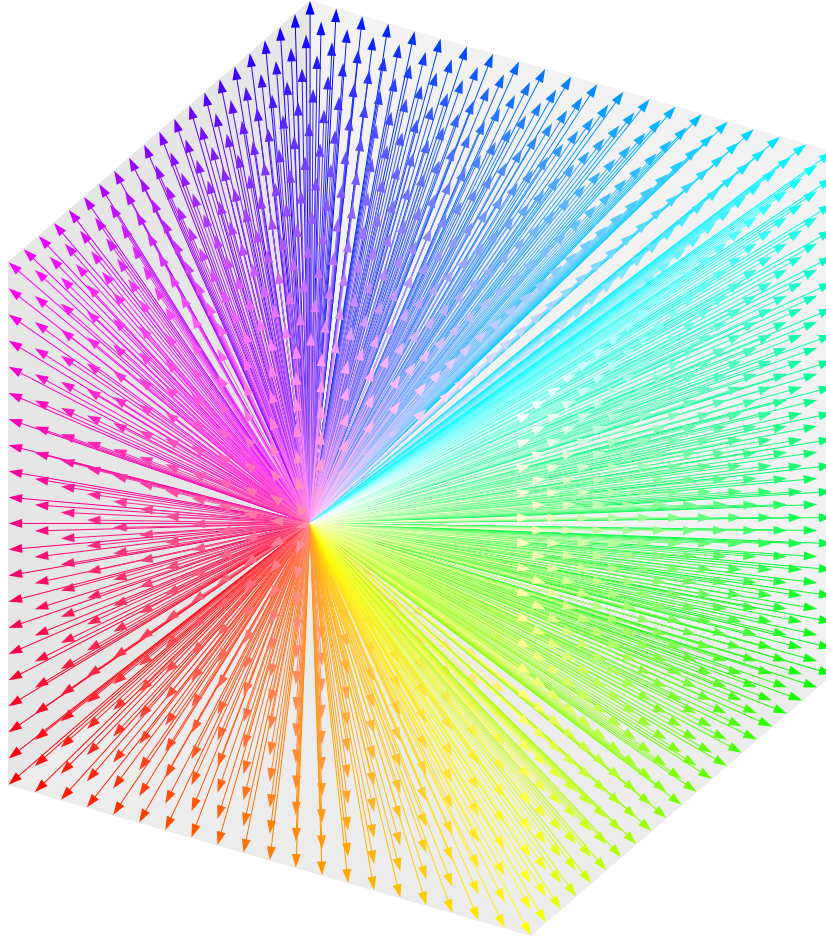


Figure 8. The Closer the Angle Between Color Vectors Is to 0° , the More Similar the Two Colors Are

The Vector Inner Product in Polar Coordinates

The polar coordinates introduced in the previous section offer an intuitive geometric explanation when deriving the vector inner product. We can represent vectors in polar coordinates and derive the inner-product formula from trigonometric relationships.

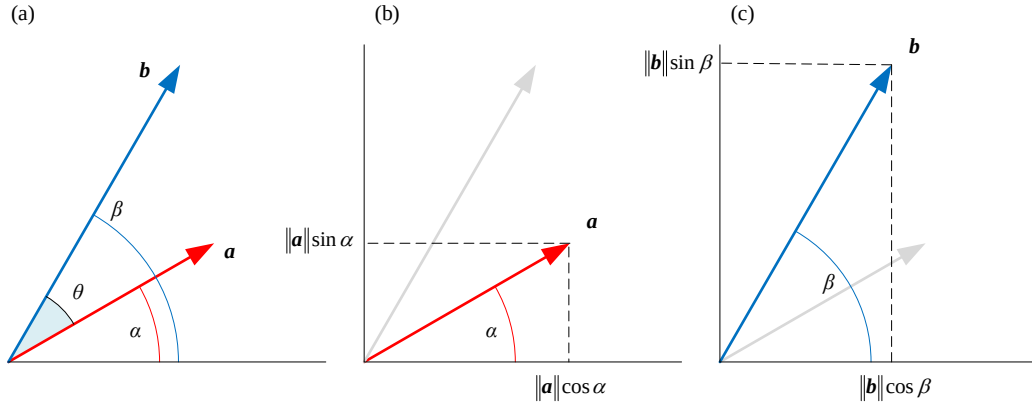
Figure 9 (a) shows the polar coordinates in which $\mathbf{a}$ and $\mathbf{b}$ can be written as the polar coordinates $(\|\mathbf{a}\|, \alpha)$ and $(\|\mathbf{b}\|, \beta)$.

α is the angle between $\mathbf{a}$ and the polar axis (the positive x_1 direction); β is the angle between $\mathbf{b}$ and the polar axis.

As shown in Figure 9 (a), the angle between $\mathbf{a}$ and $\mathbf{b}$ is $\theta = \beta - \alpha$.

By a trigonometric identity, the cosine of the angle between $\mathbf{a}$ and $\mathbf{b}$ can be expanded as

$$\cos \theta = \cos(\beta - \alpha) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

Figure 9. Polar and Cartesian Views of Vectors $\mathbf{a}$ and $\mathbf{b}$

In the Cartesian coordinate system, as shown in Figure 9 (b), the vector $\mathbf{a}$ has coordinates $(\|\mathbf{a}\|\cos\alpha, \|\mathbf{a}\|\sin\alpha)$; as shown in Figure 9 (c), the vector $\mathbf{b}$ has coordinates $(\|\mathbf{b}\|\cos\beta, \|\mathbf{b}\|\sin\beta)$.

Based on what we covered earlier in this chapter, $\mathbf{a}$ and $\mathbf{b}$ can be written in the form “length $\times$ direction vector”:

$$\mathbf{a} = \|\mathbf{a}\| \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix}, \quad \mathbf{b} = \|\mathbf{b}\| \begin{bmatrix} \cos\beta \\ \sin\beta \end{bmatrix}$$

$\mathbf{a}$ and $\mathbf{b}$ have an inner product of

$$\begin{aligned} \mathbf{a} \cdot \mathbf{b} &= \|\mathbf{a}\| \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix} \cdot \|\mathbf{b}\| \begin{bmatrix} \cos\beta \\ \sin\beta \end{bmatrix} \\ &= \|\mathbf{a}\| \|\mathbf{b}\| \begin{bmatrix} \cos\alpha \\ \sin\alpha \end{bmatrix} \cdot \begin{bmatrix} \cos\beta \\ \sin\beta \end{bmatrix} \\ &= \|\mathbf{a}\| \|\mathbf{b}\| (\cos\alpha \cos\beta + \sin\alpha \sin\beta) \\ &= \|\mathbf{a}\| \|\mathbf{b}\| \cos(\beta - \alpha) = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta \end{aligned}$$

The Law of Cosines

We can also derive the vector inner product formula using the law of cosines for triangles.

As shown in Figure 10 (b), geometrically, the law of cosines for a triangle is

$$c^2 = a^2 + b^2 - 2ab \cos\theta$$

The triangle's side lengths a, b, c correspond respectively to the lengths of vectors $\mathbf{a}, \mathbf{b}$, and $\mathbf{c}$.

The equation above can be written as

$$\|\mathbf{c}\|^2 = \|\mathbf{a}\|^2 + \|\mathbf{b}\|^2 - 2\|\mathbf{a}\|\|\mathbf{b}\|\cos\theta$$

As shown in Figure 10 (a), introduce the vector difference $\mathbf{c} = \mathbf{a} - \mathbf{b}$, the square of the length of $\mathbf{c}$ is

$$\begin{aligned}
 \|c\|^2 &= \|a-b\|^2 \\
 &= (a-b) \cdot (a-b) \\
 &= a \cdot a + b \cdot b - 2a \cdot b \\
 &= \|a\|^2 + \|b\|^2 - 2a \cdot b
 \end{aligned}$$

Comparing the two equations above, we readily obtain $\|a\|\|b\|\cos\theta = a \cdot b$.

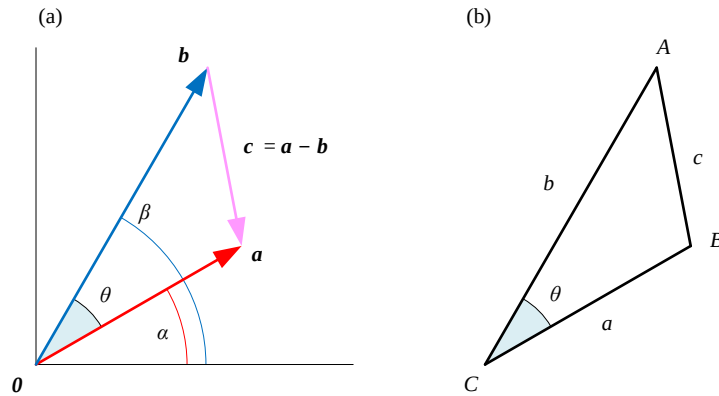


Figure 10. The Law of Cosines and the Vector Inner Product



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Compute the inner product and angle for each of the following vector pairs, and explain the geometric meaning.

- $[1, 0]^T, [0, 1]^T$;
- $[-1, 0]^T, [0, -1]^T$;
- $[1, 0]^T, [-1, 0]^T$;
- $[3, 4]^T, [-4, 3]^T$;

Q2. Given $x = [x_1, x_2]^T$ and $[1, 1]^T$ are orthogonal, find the relationship between x_1 and x_2 that satisfies this condition, and try to describe what this relationship represents geometrically.

Q3. Given $x = [x_1, x_2, x_3]^T$ and $[1, 1, 1]^T$ are orthogonal, find the relationship between x_1, x_2 , and x_3 that satisfies this condition, and try to describe what this relationship represents geometrically.

Q4. Write Python code to compute the inner product, cosine of the angle, and angle for each vector pair in Figure 6.

Q5. Use Python to generate two 3-dimensional random integer vectors, with components in the range $[0, 9]$. Compute and print the dot product and angle between the two vectors.

Q6. Study on your own how to compute the vector cross product.

<https://mathworld.wolfram.com/CrossProduct.html>

Q7. Study cosine similarity on your own — the cosine of the angle between two vectors, widely used in machine learning and search to measure how similar two vectors (e.g., text or image embeddings) are, regardless of their length.

Q8. Preview scalar projection and vector projection.

Q9. Study the Hadamard product (element-wise product) of vectors on your own.